

Two resolutions over 46 seconds

What the extended 256^3 run settles, and the one thing it unsettles

Report II — companion to `report_rot192_rot256`

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Abstract

The first report compared the 192^3 and 256^3 runs at $t = 12$ s. The 256^3 run has since been carried to $t = 58.2$ s, so the comparison now spans 46 seconds rather than three. Three results follow. **The braking of L_z converges:** three successive regimes, each about half the rate of the one before, agreeing between the grids to 7, 7 and 15 percent — so the identification of the angular-momentum transport as magnetic, previously resting on one resolution, now rests on two. **The steepening of the differential rotation converges,** 22.6% against 20.3% over the full baseline. **The residual magnetic field does not:** it settles a factor of 25 higher on the finer grid and is still rising at $t = 58$ s. A methodological result comes with the second: over a 14 s window the same two runs appear to disagree about the steepening by a factor of eight, because the 256^3 onset is some 6 s late. Two earlier drafts drew two different wrong conclusions from that window.

1 What this report adds

Report I established the configuration, the instability and the survival of the star, and compared the two grids over the three seconds where both existed. Its convergence statements were therefore all short-baseline statements. Since then the 256^3 run has been continued through twelve chained windows to $t = 58.2$ s and all 164 plotfiles have been processed with the same diagnostics, so every quantity in Report I can now be asked of both grids over almost the whole evolution.

Nothing about the configuration, the code or the diagnostics has changed. The star is the same $2.005 M_\odot$ differentially rotating equilibrium with a toroidal-dominated field; the tool is the same `tools/fbtbp.cpp` rebuilding everything from B_x, B_y, B_z .

Throughout, comparisons are made between means over two pulsation periods at each end of the baseline, never between single snapshots. The star breathes by $\pm 14\%$ every 1.498 s, and a snapshot catches an arbitrary phase of it.

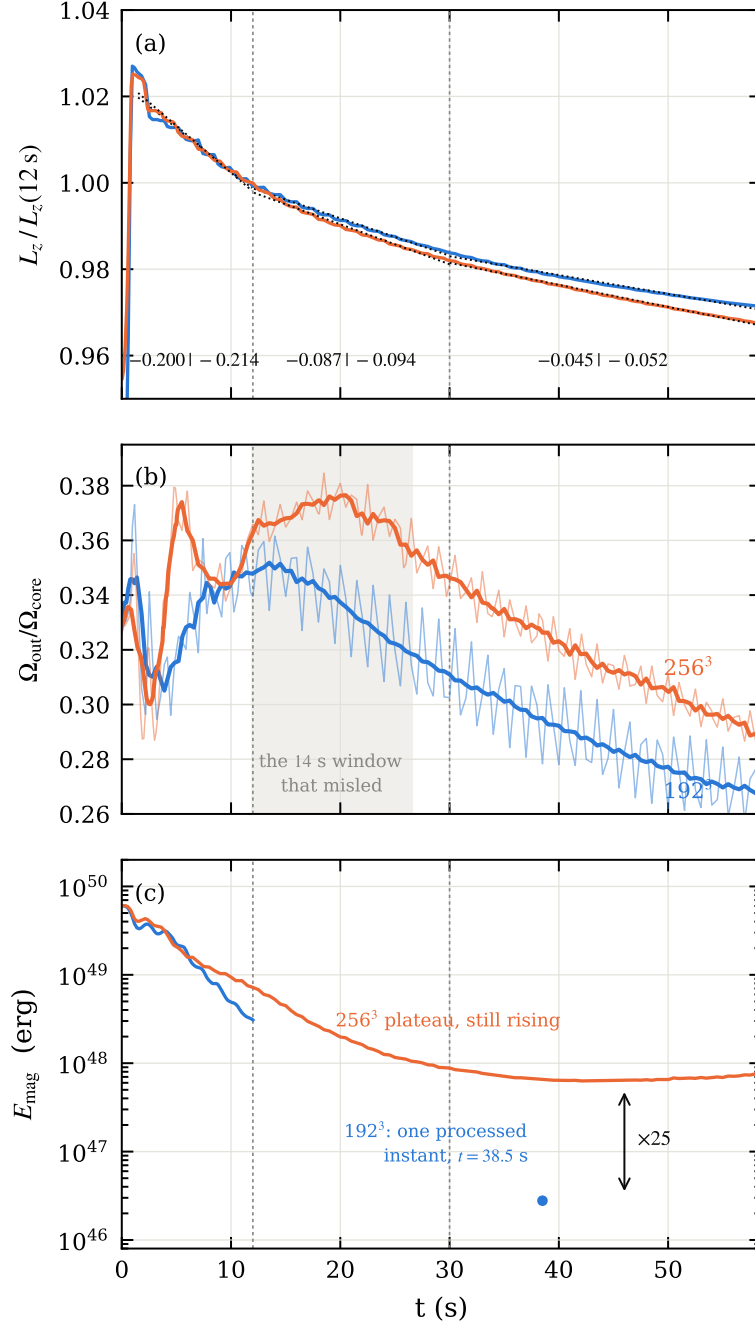


Figure 1: The three results. **(a)** L_z normalised at $t = 12$ s, with the three braking regimes fitted; the numbers along the bottom are $192^3 | 256^3$ in percent per second, and the dashed verticals are the regime boundaries. **(b)** The differential rotation. The finer grid begins to steepen about 6 s late, then steepens faster; the shaded band is the 14 s window over which that lag reads as a disagreement. **(c)** Magnetic energy, the one panel that does not converge. The 192^3 field diagnostics stop at $t = 12$ s; the isolated point at $t = 38.5$ s is the single late instant that was processed, and is drawn as a marker because that is all it is.

2 The field, before and after

Before the numbers, the pictures. Fig. 2 is what the ordered field does at 192^3 : axisymmetric circles of B_ϕ in the equatorial cut at $t = 0$, an $m = 1$ deformation growing through $t \simeq 2$ –4 s, filamentation at saturation, and a disordered remnant afterwards. The poloidal row below it starts almost empty — the initial poloidal field is three decades below the toroidal one — and fills in as the instability converts one into the other.

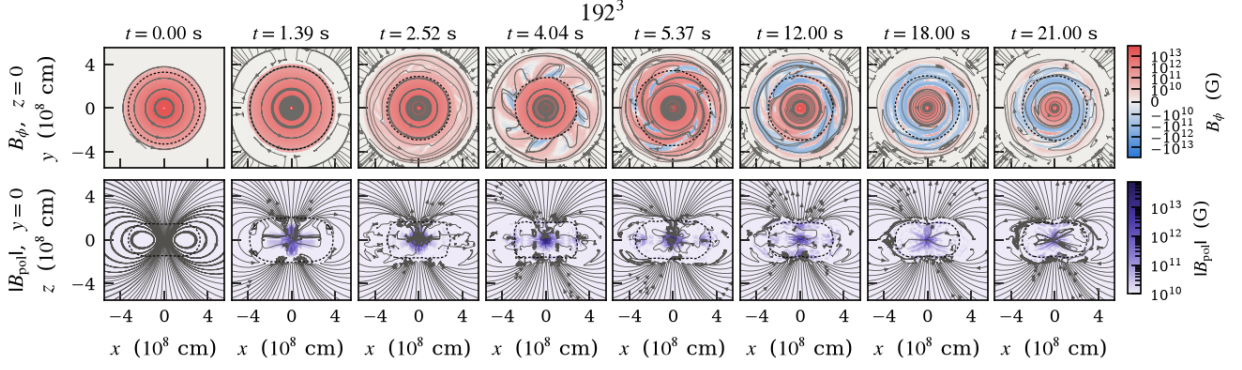


Figure 2: 192^3 , $t = 0$ to 21 s. **Top:** equatorial cut $z = 0$, where B_ϕ lies in the plane, so the streamlines are the toroidal field lines — circles while the star is axisymmetric, and their departure from circles is the instability. **Bottom:** meridional cut $y = 0$, where the in-plane field is the poloidal one. Colour is B_ϕ and $|B_{\text{pol}}|$, both starting at 10^{10} G so the two rows share a threshold in gauss. The dashed contour is $\rho = 10^6 \text{ g cm}^{-3}$.

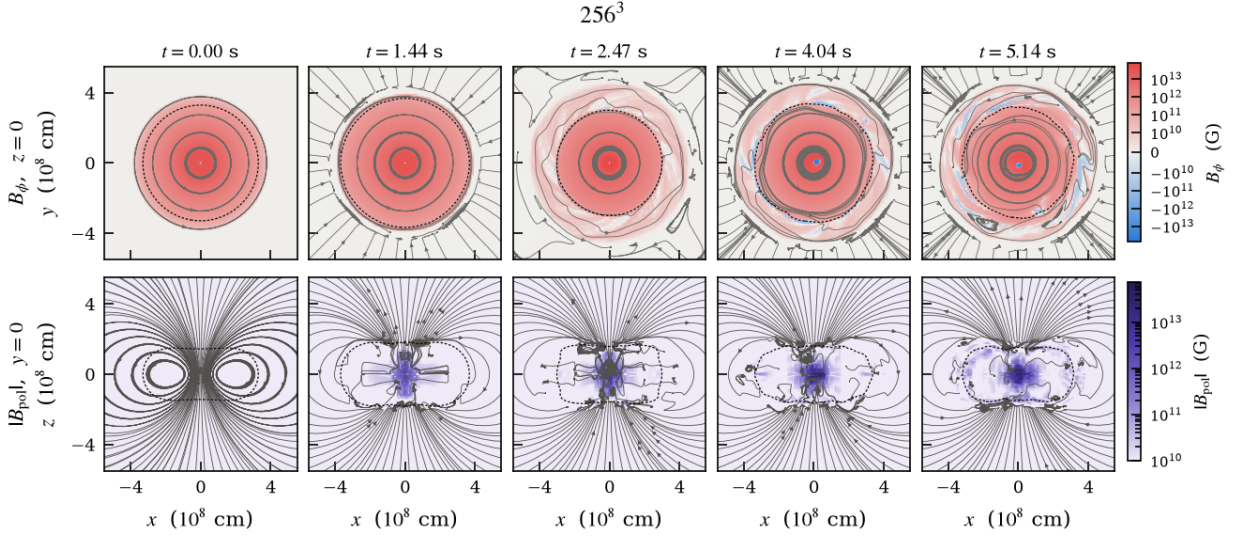


Figure 3: 256^3 , same planes, same colour scales, same threshold as Fig. 2 — the two are meant to be compared panel by panel. **This sequence stops at $t = 5.14$ s**, the end of the growth phase, because those are the only instants for which slices have been extracted. It therefore covers none of the window this report is about. Extending it is the obvious companion to Sec. 5: a slice at $t \simeq 58$ s would show directly whether the residual field that fails to converge is organised or turbulent, which is the physical question underneath the two readings offered there.

The comparison that matters for this report is not visible in either figure. Both stop before or barely into the window where the two grids disagree about the residual, and the disagreement is a factor of 25 in energy — large enough that it should be plainly visible in a slice, as either a different field strength or a different degree of organisation. That figure is the one piece of evidence this report is missing.

3 The braking of L_z converges

This is the strongest result of the comparison, and it is the one that matters most for the physics, because it is what identifies the transport.

There is no explicit viscosity and no explicit resistivity in these runs, so nothing else in the scheme brakes anything. The star loses angular momentum twice as fast while the ordered toroidal

$d \ln L_z / dt$		192 ³	256 ³	agreement
$t = 1.5\text{--}11.9\text{ s}$	toroidal field alive	$-0.1996\%/s$	$-0.2144\%/s$	7%
$t = 12\text{--}30\text{ s}$	field decaying	$-0.0873\%/s$	$-0.0935\%/s$	7%
$t = 30\text{--}58\text{ s}$	residual	$-0.0446\%/s$	$-0.0523\%/s$	15%

Table 1: Three regimes, each braking at about half the rate of the one before, and the two grids agree on all three.

field exists as it does afterwards, and half as fast again once even the decaying field has gone. That factor of two, repeated, is the signature of magnetic transport, and it is now measured at two resolutions instead of one.

The third regime is new here — Report I described two. With the baseline extended it is clear that the braking does not simply switch from “fast” to “slow” when the field dies but continues to weaken as the field weakens, which is what a magnetic mechanism should do and what a numerical one need not.

What this does not establish. The rates agree between the grids; that does not make them physical. Both runs brake through whatever transport the scheme supplies, and if a component of that is numerical it may well converge at the same rate as the physical part. The argument for the magnetic identification is the *correlation with the field*, not the convergence; convergence only removes the worry that the correlation was a coarse-grid accident.

4 The steepening converges — over a long enough baseline

The differential rotation is the property the configuration depends on: a $2 M_\odot$ white dwarf exists here only because Ω falls steeply outwards. Report I found that the profile not only survives but steepens, and flagged the amplitude as unconverged.

12–15 s $\rightarrow$ 55–58.3 s	192 ³	256 ³	
L_z	-2.6%	-2.9%	converged
Ω_{out}	-15.3%	-14.9%	converged (3%)
$\Omega_{\text{out}}/\Omega_{\text{core}}$	-22.6%	-20.3%	converged (10%)
Ω_{core}	$+9.4\%$	$+6.7\%$	same sign
$L_z^{\text{outer}}/L_z^{\text{inner}}$	-10.6%	-6.2%	same sign

Table 2: The steepening over 46 s. The amplitude converges to 10% and the outer-shell spin-down to 3%.

4.1 A convergence test shorter than the lag measures the lag

The amplitude looked irreducibly resolution-dependent over shorter windows, and the reason is worth recording because it is a trap that will recur in this problem.

Neither number is wrong. The short baseline simply does not contain the onset, and a delayed effect measured over a window shorter than its delay is indistinguishable from an absent effect — and, if the lagging curve happens to drift the other way first, from an effect of the opposite sign.

This cost two wrong conclusions in successive drafts of Report I: first that the steepening was a coarse-grid artefact, then that its direction was real but its amplitude was not. Both were stated with the numbers that supported them, and both were wrong for the same reason.

The general form is uncomfortable, because the lag is not known in advance and is not obviously bounded. Here it is 6 s against a 1.5 s pulsation and a 2.4 s Alfvén time — a few dynamical times,

change in $\Omega_{\text{out}}/\Omega_{\text{core}}$	192 ³	256 ³
$t = 12 \rightarrow 18$ s	−3.2%	+ 2.1%
$t = 18 \rightarrow 24$ s	−5.1%	−3.0%
$t = 24 \rightarrow 30$ s	−4.5%	− 5.6%
$t = 30 \rightarrow 38$ s	−3.8%	− 5.2%
total over a 14 s baseline	−8.2%	−1.0%
total over a 46 s baseline	−22.6%	−20.3%

Table 3: The finer grid is still flat — rising, even — while the coarse grid has already begun, and then steepens faster. The last two rows are the same comparison over two baselines: a factor of eight apart at 14 s, ten percent apart at 46 s.

which is reassuring but is a measurement, not a principle. Any future convergence test in this problem needs a baseline long enough that a plausible lag is a small fraction of it.

5 The residual field does not converge

The one place where extending the baseline made the disagreement worse.

late-time field	192 ³	256 ³	ratio
E_{mag} plateau (erg)	$\simeq 3 \times 10^{46}$	7.0×10^{47}	25
max $ B $ (G)	$\simeq 1 \times 10^{12}$	6×10^{12}	6
trend for $t > 45$ s	—	+ 0.012 s ^{−1}	<i>rising</i>

Table 4: The 192³ values are from the single late instant processed with the field diagnostics, $t = 38.5$ s; the 256³ values are means over $t = 54$ –58.3 s. The trend is fitted to $\ln E_{\text{mag}}$ over $t > 45$ s, with E_{tor} and E_{pol} both rising.

This is consistent with what Report I already found: with no explicit resistivity, the decay of E_{tor} is the numerical resistivity, and it scales as $dx^{2.4}$. Less numerical dissipation leaves more field. But the consequences are sharper than the factor-of-two disagreement seen at $t = 12$ s:

- The “disordered residual of order 10^{12} G” reported at 192³ describes a floor set by the mesh, not a state of the star.
- The factor 2170 by which E_{mag} falls at 192³ is an **upper bound** on the destruction, and not a tight one.
- The 256³ run has not reached a final state at all. A run that is gaining magnetic energy when it stops has not finished.

Two readings, and two resolutions cannot separate them. The mechanism for genuine regeneration is available and it is not exotic: the differential rotation survives and is still steepening, so there is free shear energy, and winding residual poloidal field into toroidal is the most ordinary thing a shear flow can do to a magnetic field. On that reading the finer grid is showing the beginning of a shear dynamo that the coarse grid dissipates before it can start.

The other reading is that both runs are on the same decaying track and the finer one is simply further from the mesh floor, with the apparent rise being the last decade of a decay it cannot resolve, contaminated by the pulsation.

The two look identical in the data we have. Distinguishing them needs either a third resolution — 384³, about five times the cost of 256³ — or, better, explicit resistivity, which would give the

problem a physical dissipation scale instead of letting the mesh set it. CASTRO does not currently implement it.

6 What changes in Report I

1. **Strengthened.** The magnetic identification of the transport. Two grids, three regimes, agreement of 7–15%.
2. **Strengthened.** The steepening of the differential rotation, now converged in amplitude as well as direction.
3. **Weakened.** Everything about the end state of the field. The residual is a mesh floor and the destruction factor is an upper bound. Report I’s Table 4 should be read with that in mind: its mass, radii and rotation rows are measurements, its field rows are bounds.
4. **New caveat.** Convergence tests in this problem need baselines longer than the inter-grid lag, and the lag is only known afterwards.

Status and next steps

192³: complete, $t = 60$ s, field diagnostics processed to $t = 12$ s plus the single instant at 38.5 s.
256³: $t = 58.2$ s, all 164 plotfiles processed. A final window to 60 s is queued; nothing here depends on it.

The cheapest thing that would sharpen this report is not more physical time but the missing 192³ field data: processing its plotfiles between $t = 12$ and 60 s would turn the single point in Fig. 1c into a curve and make the residual comparison a proper one rather than a snapshot against a mean. It costs no new simulation.

Beyond that, the run that would test the interpretation directly is a genuinely mixed field — comparable poloidal and toroidal energy, rather than the 10^{-7} ratio used here. A toroidal-dominated field is the textbook unstable case and behaved like one; the twisted-torus configuration is the one expected to survive, and it is the geometry the equilibrium literature on super-Chandrasekhar white dwarfs does not construct.

Inputs, submission scripts, diagnostic tools, data and figure code are under version control in the WD_MAG repository. The data behind every number here is `investigations/bt_bp_256_long.csv` (164 rows, $t = 0$ to 58.2 s) and `investigations/rotation_192.csv`.